Supporting Information

Efficient Fischer-Tropsch to light olefins over iron-based catalyst with low methane selectivity and high olefin/paraffin ratio

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Detailed description about materials syntheses

Support SP1 and catalyst N1: The nominal composition of support SP1 is 100% ($ZnAl_2O_4$•$1Al_2O_3$). Support SP1 is synthesized by coprecipitation. Zinc nitrate and aluminum nitrate are employed, while the dosages are according to the ratios of elemental compositions in Table 1. Zinc nitrate and aluminum nitrate are dissolved together in DI water to form a solution with a weight ratio of metal (i.e., Zn & Al) sources to water of 1:5. Sodium bicarbonate is dissolved in water to form a precipitant solution with a weight ratio of sodium bicarbonate to water of 1:13. Support SP1 is synthesized by dropping the metal (i.e., Zn & Al) source solution and the precipitant solution together at 40°C under stirring, filtered after 1 h aging period, washed by water to a neutral pH, half-dried at 80°C, kneaded, extruded, air-dried at RT, oven-dried at 110°C for 2 h, and then annealed at 1200°C for 24 h in air. Above is the procedure for support SP1. The nominal composition of catalyst N1 is 15%Fe/2%K_2O/83%($ZnAl_2O_4$•$1Al_2O_3$). Unreduced catalyst N1 is synthesized by incipient wetness of support SP1 with a K_2CO_3 solution, desiccation at 110°C for 2 h, another incipient wetness with an ammonium iron citrate solution, another desiccation at 110°C for 2 h, and then anneal at 350°C for 4 h in air. Reduced catalyst N1 is prepared by a subsequent reduction at 550°C for 6 h in an atmosphere of 50%H_2/50%Ar. Generally, K_2O% is employed to simplify the expression of potassium content in the catalyst composition of potassium carbonate application.

Support SP5 and catalyst N5: The nominal composition of support SP5 is 100% ($ZnAl_2O_4$•$5Al_2O_3$). Support SP5 is synthesized by coprecipitation. Zinc nitrate and aluminum nitrate are dissolved together in DI water to form a solution with a weight ratio of metal (i.e., Zn & Al) sources to water of 1:5. Sodium bicarbonate is dissolved in H_2O to form a precipitant solution with a weight ratio of sodium bicarbonate to water of 1:13. Support SP5 is synthesized by dropping the metal (i.e., Zn & Al) source solution and the precipitant solution together at 40°C under stirring, filtered after 1 h aging period, washed by water to a neutral pH, half-dried at 80°C, kneaded, extruded, air-dried at RT, oven-dried at 110°C for 2 h, and then annealed at 1200°C for 24 h in air.

Above is the procedure for support SP5. The nominal composition of catalyst N5 is 15%Fe/2%K_2O/83%($ZnAl_2O_4$•5Al_2O_3). Unreduced catalyst N5 is synthesized by incipient wetness of support SP5 with a K_2CO_3 solution, desiccation at 110°C for 2 h, another incipient wetness with an ammonium iron citrate solution, another desiccation at 110°C for 2 h, and then anneal at 350°C for 4 h in air. Reduced catalyst N5 is prepared by a subsequent reduction at 550°C for 6 h in an atmosphere of 50%H_2/50%Ar.

Support SP∞ and catalyst REF∞: The nominal composition of support SP∞ is 100%(Al_2O_3). Support SP∞ is synthesized by precipitation. Aluminum nitrate is employed is dissolved in H_2O to form a solution with a weight ratio of metal (i.e., Al) source to DI water of 1:5. Sodium bicarbonate is dissolved in water to form a precipitant solution with a weight ratio of sodium bicarbonate to water of 1:13. Support SP∞ is synthesized by dropping the metal (i.e., Al) source solution and the precipitant solution together at 40°C under stirring, filtered after 1 h aging period, washed by water to a neutral pH, half-dried at 80°C, kneaded, extruded, air-dried at RT, oven-dried at 110°C for 2 h, and then annealed at 1200°C for 24 h in air. Above is the procedure for support SP∞. The nominal composition of catalyst REF∞ is 15%Fe/2%K_2O/83%(Al_2O_3). Unreduced catalyst REF∞ is synthesized by incipient wetness of support SP∞with a K_2CO_3 solution, desiccation at 110°C for 2 h, another incipient wetness with an ammonium iron citrate solution, another desiccation at 110°C for 2 h, and then anneal at 350°C for 4 h in air. Reduced catalyst REF∞ is synthesized by a subsequent reduction at 550°C for 6 h in an atmosphere of 50%H_2/50%Ar.

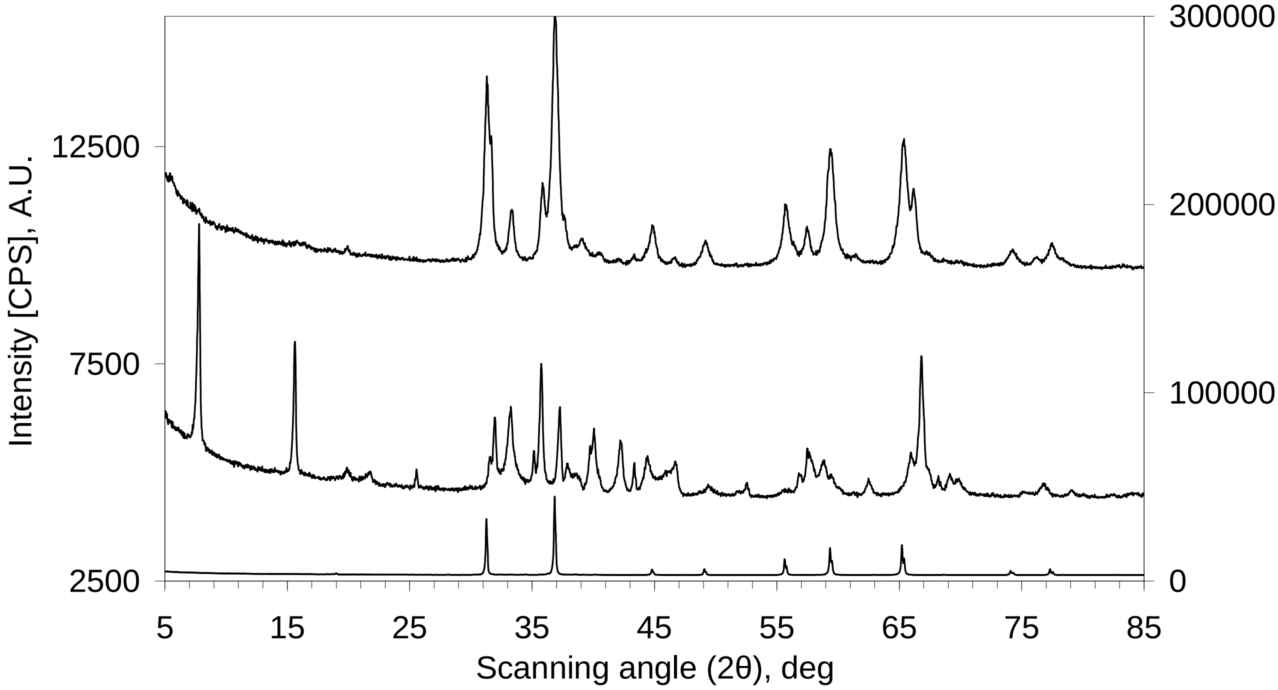
Support SP0 and catalyst REF0: The nominal composition of support SP0 is 100%($ZnAl_2O_4$). Support SP0 is synthesized by coprecipitation. Zinc nitrate and aluminum nitrate are employed, while the dosages are according to the ratios of elemental compositions in Table 1. Zinc nitrate and aluminum nitrate are dissolved together in DI water to form a solution with a weight ratio of metal (i.e., Zn & Al) sources to water of 1:5. Sodium bicarbonate is dissolved in water to form a precipitant solution with a weight ratio of sodium bicarbonate to water of 1:13. Support SP0 is synthesized by dropping the metal (i.e., Zn & Al) source solution and the precipitant solution together at 40°C under stirring, filtered after 1 h aging period, washed by water to a neutral pH, half-dried at 80°C, kneaded, extruded, air-dried at RT, oven-dried at 110°C for 2 h, and then annealed at 1200°C for 24 h in air. Above is the procedure for support SP0. The nominal composition of catalyst REF0 is 15%Fe/2%K_2O/83%($ZnAl_2O_4$). Unreduced catalyst REF0 is synthesized by incipient wetness of support SP0 with a K_2CO_3 solution, desiccation at 110°C for 2 h, another incipient wetness with an ammonium iron citrate solution, another desiccation at 110°C for 2 h, and then anneal at 350°C for 4 h in air. Reduced catalyst REF0 is prepared by a subsequent reduction at 550°C for 6 h in an atmosphere of 50%H_2/50%Ar.

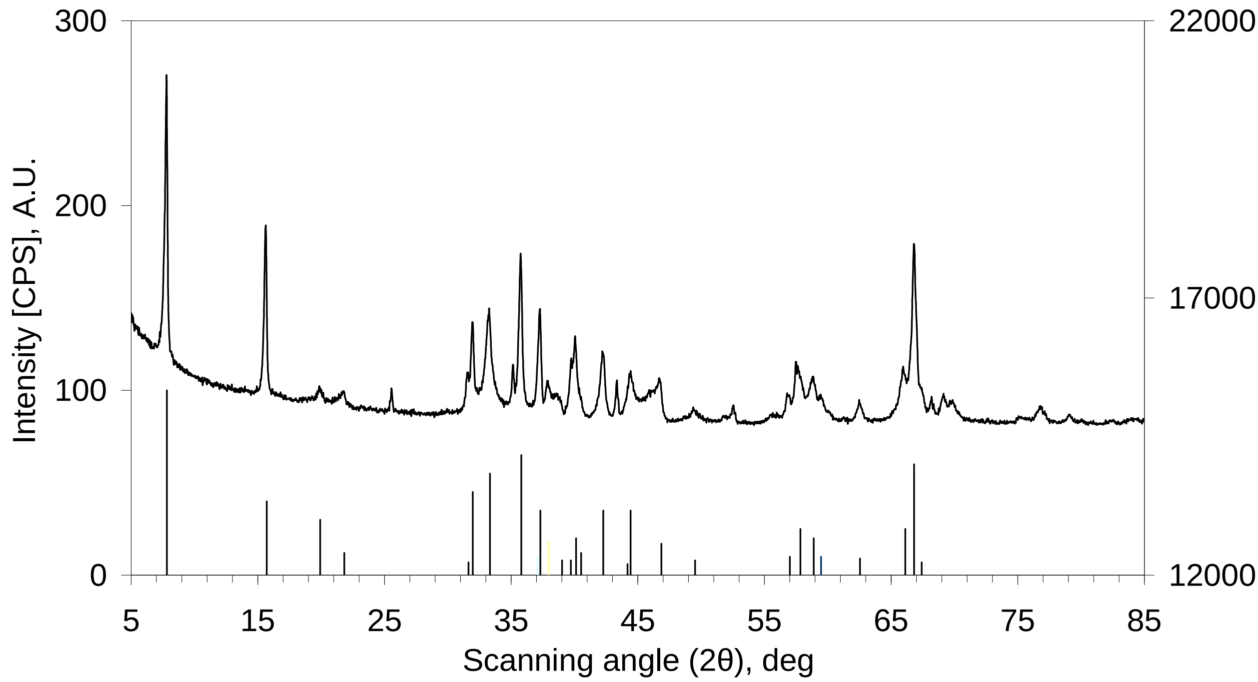
Characterization of materials

The structure analysis of materials are performed with powder X-ray diffraction (XRD) using a Panalytical X'Pert Pro diffractometer with Cu Kα radiation. The morphologies of materials are observed with Field-emission scanning electron microscopy (FESEM) using a JSM-7001F FESEM. Elemental contents were measured with an Agilent 725 ICP-OES. The mesoporosity property is analyzed with nitrogen physisorption using a Quantachrome NOVA1000 surface area and pore size analyzer. Surface basicity of materials is measured with CO_2-TPD (CO_2 temperature-programmed desorption) using a Quantachrome ChemStar™ chemisorption analyzer (Detector: Chem-Bet). Samples for CO_2-TPD tests are loaded, *in-situ* reduced at 550°C and then cooled down to 50°C. Adsorption of CO_2 is conducted at 50°C with a mixed gas of 2000 ppm CO_2 in Ar, followed by helium flushing at 50°C for the removal of physisorbed CO_2. The CO_2-TPD data were collected in the temperature range of 50～800°C at atmospheric pressure. The weight of each CO_2-TPD sample is 0.1000 g.

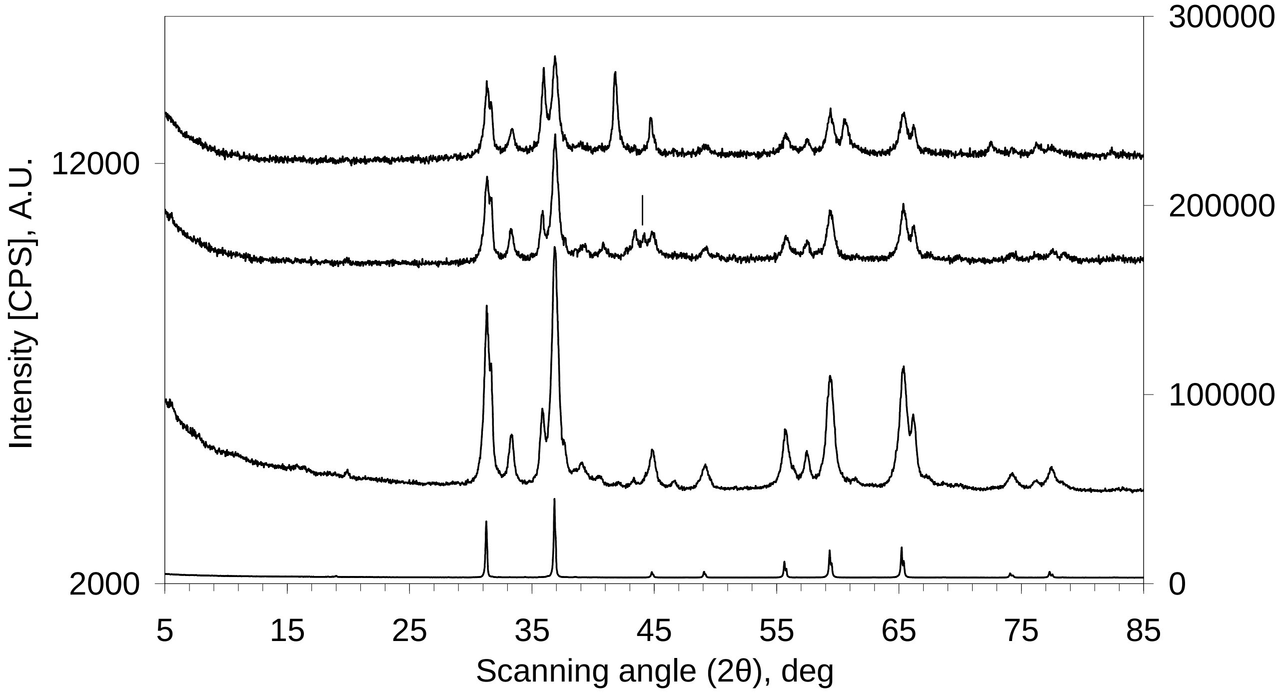
Catalytic Performance Tests

Catalytic CO hydrogenation reactions are carried out with a fixed-bed tubular reactor. Catalysts (i.e., in oxide status, 40-60 mesh) are loaded and then *in-situ* reduced at 550°C for 6 h by 50%H_2/50%Ar at 0.1 MPa. A pre-mixed syngas (H_2/CO/Ar = 45/45/10, by mole%) is applied over the *in-situ* reduced catalysts for CO hydrogenation reaction at a total pressure of 2.0 MPa as well as a space velocity (SV) of 1500 mL•g cat^{-1}•h^{-1}. Syngas passes through the reactor in one way. Catalytic data are determined after the reactions reach steady state. Reaction temperature is changed from low to high during catalytic tests. UHP argon is utilized as the internal standard of syngas in GC analysis. CO, CH_4 and CO_2 are analyzed using a GC with a TCD detector equipped with a packed column (TDX-01, 6 m in length). Incondensable gaseous hydrocarbons are analyzed using a GC with a FID detector equipped with a capillary column (PLOT Al_2O_3/S, 50 m * 0.53 mm * 25 μm). Water phase is analyzed by a GC with a TCD detector equipped with a packed column (GDX-102, 2 m * 3 mm). Oxygenates in water phase are negligible and thus not counted in calculations. Condensed wax and oil (dissolved in cyclohexane) are analyzed by a GC with a FID detector equipped with a capillary column (OV-101, 50 m * 0.25 mm * 0.5 μm).

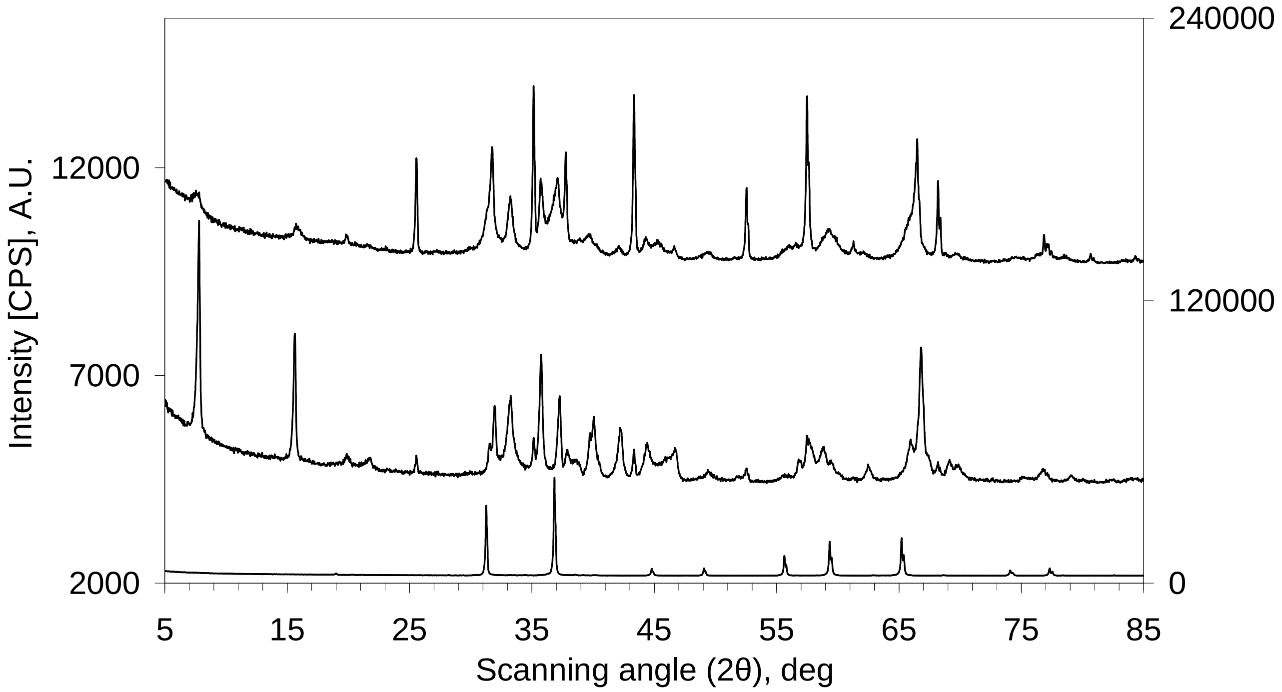




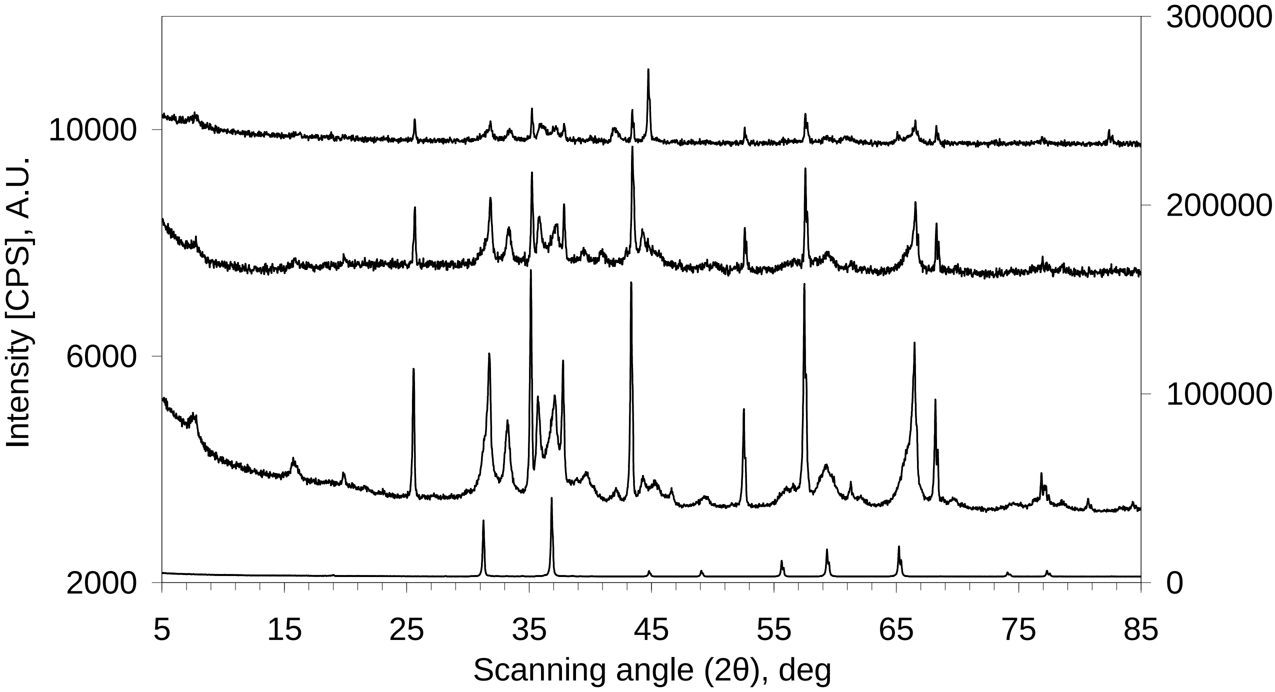
Supporting Information Fig.S1. Enlarged X-ray diffraction (XRD) patterns of some support samples. SP1: 100% ($ZnAl_2O_4$•$1Al_2O_3$). SP∞: 100%(Al_2O_3). SP0: 100%($ZnAl_2O_4$). a): Pattern "SP0" follows the scale of the right ordinate axis, while others follow the left ordinate axis. b): Pattern "SP∞" follows the scale of the right ordinate axis. The column curve shows the standard diffraction lines of β-Al_2O_3 (i.e., $NaAl_{11}O_{17}$ or $Na_2Al_{22}O_{34}$, JCPDS No. 31-1263).



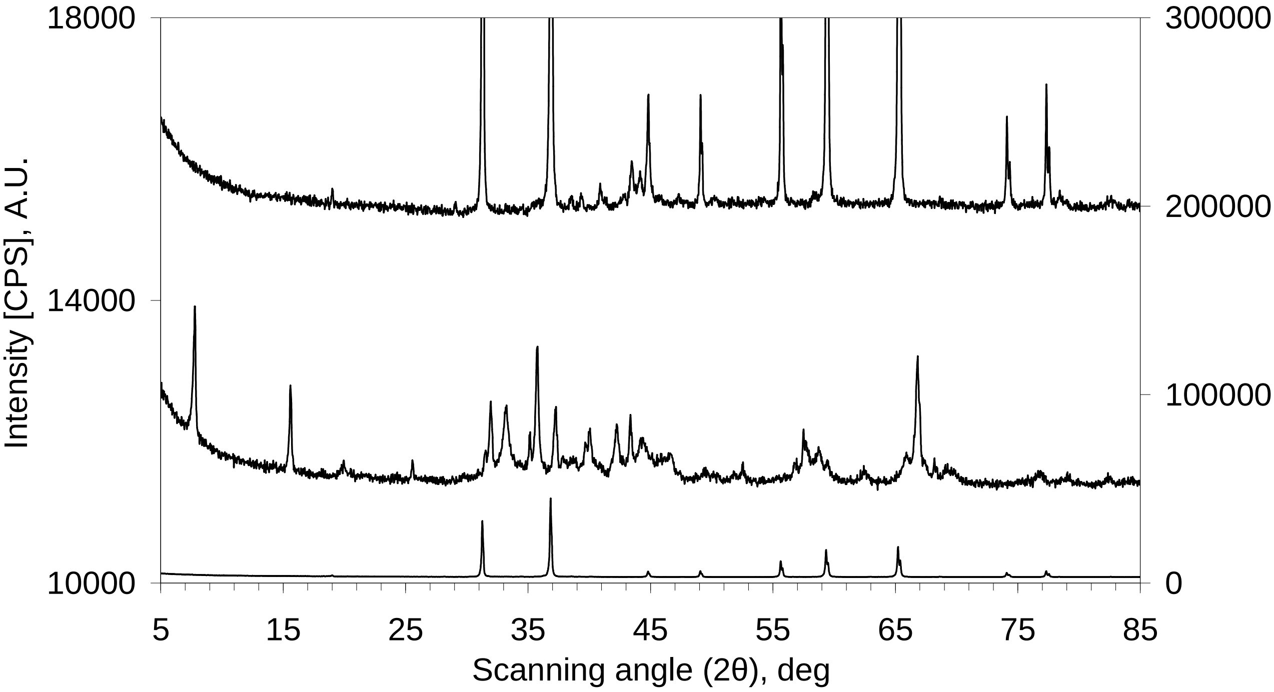
Supporting Information Fig.S2. Enlarged X-ray diffraction (XRD) patterns of freshly reduced N1 and used N1. N1: 15%Fe/2%K_2O/83%($ZnAl_2O_4$•1Al_2O_3). SP1: 100%($ZnAl_2O_4$•1Al_2O_3). SP0: 100%($ZnAl_2O_4$). Pattern “SP0” follows the scale of the right ordinate axis, while others follow the left ordinate axis.



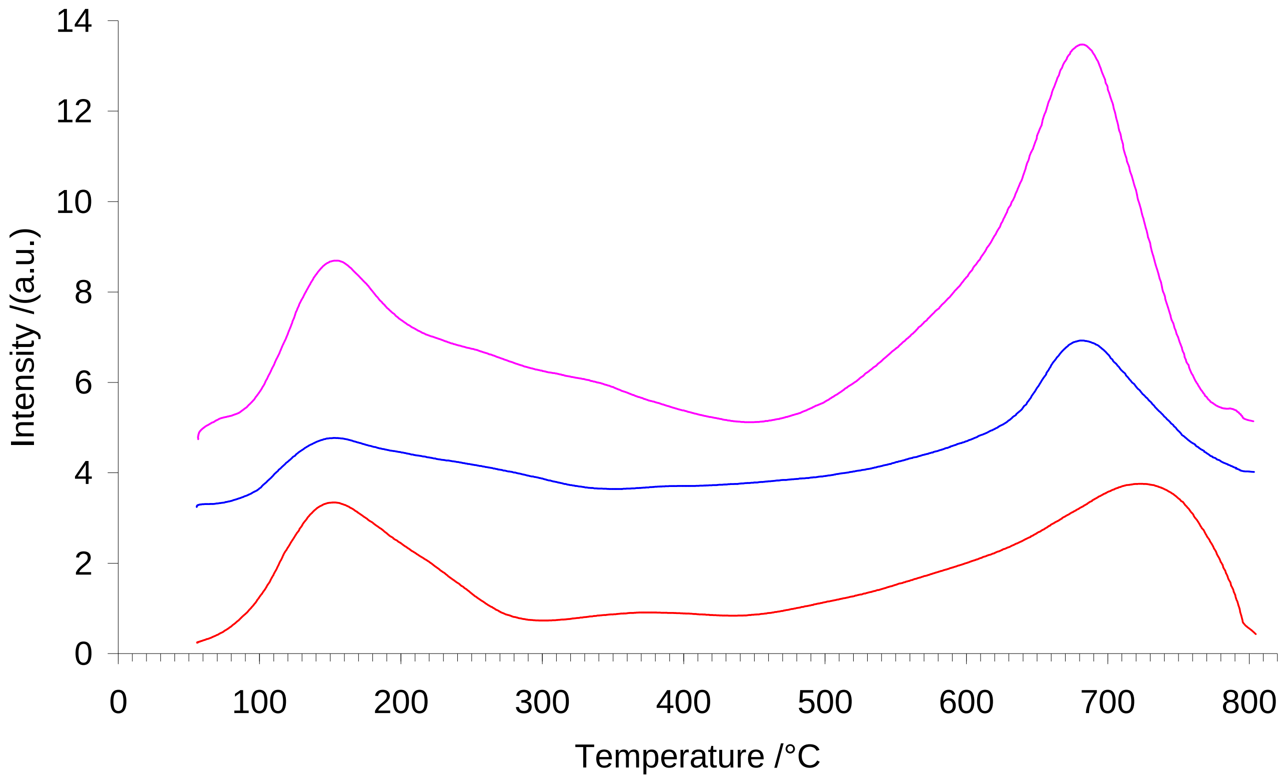
Supporting Information Fig.S3. Enlarged X-ray diffraction (XRD) patterns of some support samples. SP5: 100% ($ZnAl_2O_4$•5Al_2O_3). SP∞: 100%(Al_2O_3). SP0: 100%($ZnAl_2O_4$). Pattern “SP0” follows the scale of the right ordinate axis, while others follow the left ordinate axis.



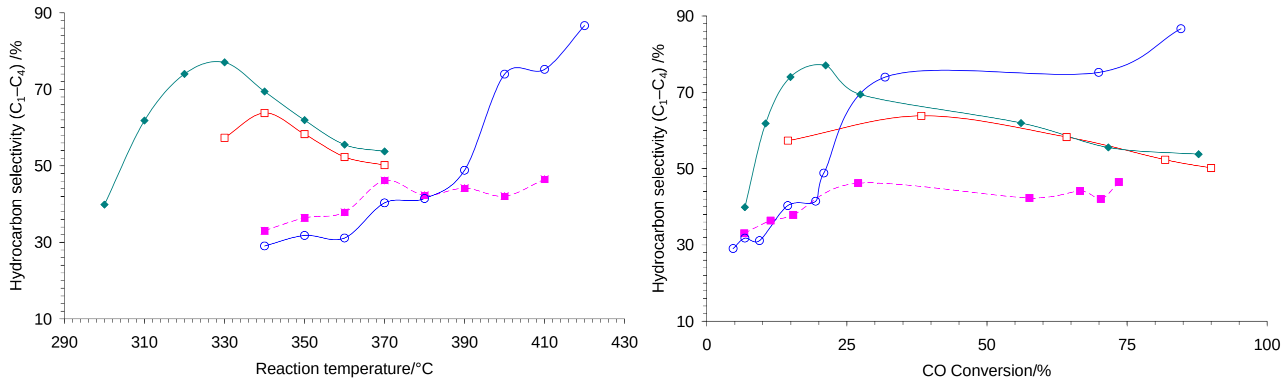
Supporting Information Fig.S4. Enlarged X-ray diffraction (XRD) patterns of freshly reduced N5 and used N5. N5: 15%Fe/2%K_2O/83%($ZnAl_2O_4$•5Al_2O_3). SP5: 100%($ZnAl_2O_4$•5Al_2O_3). SP0: 100%($ZnAl_2O_4$). Pattern "SP0" follows the scale of the right ordinate axis, while others follow the left ordinate axis.



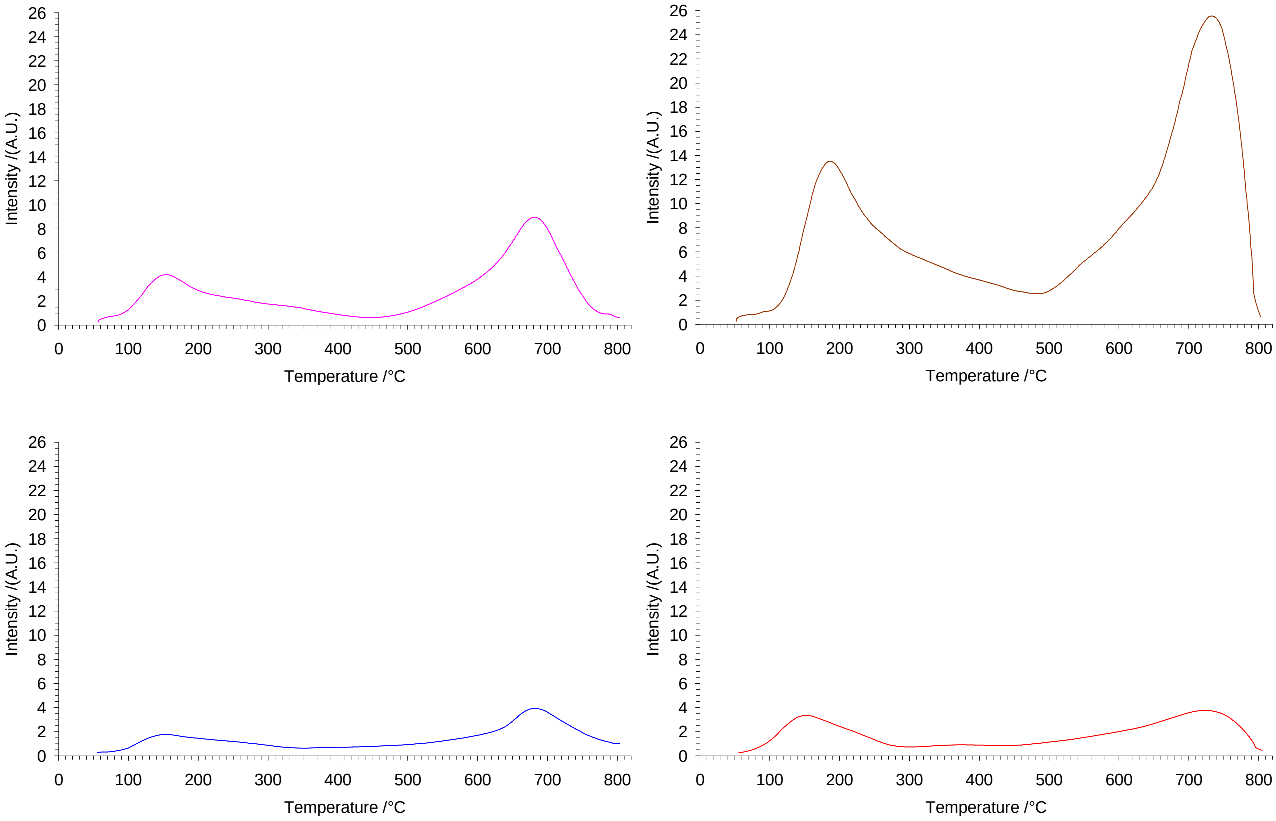
Supporting Information Fig.S5. Enlarged X-ray diffraction (XRD) patterns of used REF0 and used REF∞. REF0: 15%Fe/2%K_2O/83%($ZnAl_2O_4$). REF∞: 15%Fe/2%K_2O/83%(Al_2O_3). SP0: 100%($ZnAl_2O_4$). Pattern "SP0" follows the scale of the right ordinate axis, while others follow the left ordinate axis.



Supporting Information Fig.S6. Enlarged profiles for CO_2 temperature-programmed desorption (CO_2-TPD) of freshly reduced catalysts N1, REF0 and REF∞.



Supporting Information Fig.S7. C_1-C_4 hydrocarbon distributions (i.e., C_1-C_4 HC selectivity) of reduced catalysts at different reaction temperatures (a) and different CO conversions (b).



Supporting Information Fig.S8. Separate profiles for CO_2 temperature-programmed desorption (CO_2-TPD) of freshly reduced catalysts N1, N5, REF0 and REF∞. The weight of each CO_2-TPD sample is 0.1000 g.

Supporting Information Table S1. Contents of residue sodium (Na) in supports measured by ICP-OES

support sample (nominal composition)	SP0 100%($ZnAl_2O_4$)	SP1 100%($ZnAl_2O_4$•$1Al_2O_3$)	SP5 100%($ZnAl_2O_4$•$5Al_2O_3$)	SP∞ 100%(Al_2O_3)
ICP content of sodium (Na) by weight	1.25%	1.19%	1.21%	2.49%

Supporting Information Table S2. Integral data of CO_2-TPD results *

sample code	nominal composition (metal/promoter/support) (wt)	low temp. peak			high temp. peak			A_{total}	relative A_{total}	A_{high} /A_{low}	I_{high} /I_{low}
		temp. /°C	area%	I_{low} #	temp. /°C	area%	I_{high}				
N5	15%Fe/2%K_2O/83%($ZnAl_2O_4$•$5Al_2O_3$)	186.0	35.6%	13.5	732.4	64.4%	25.6	36394	213	1.8	1.9
N1	15%Fe/2%K_2O/83%($ZnAl_2O_4$•$1Al_2O_3$)	154.1	37.3%	4.2	681.9	62.7%	9.0	17104	100	1.7	2.1
REF0	15%Fe/2%K_2O/83%($ZnAl_2O_4$)	152.8	30.1%	1.8	680.5	69.9%	3.9	8936	52	2.3	2.2
REF∞	15%Fe/2%K_2O/83%(Al_2O_3)	152.0	33.1%	3.3	722.4	66.9%	3.8	11330	66	2.0	1.1

* A_{high} denotes the integral area of high temperature peak for CO_2 desorption, while A_{low} denotes the integral area of low temperature peak for CO_2 desorption. $A_{total} = A_{high} + A_{low}$. The relative A_{total} of N1 is specified to be 100. # Peak intensity after background deduction.

Supporting Information Table S3. Catalytic performance of catalyst N1*

T /°C	X_{CO} /%	S_{CO2} /%	hydrocarbon selectivity/% (C mole%)								olefin/paraffin ratio (O/P)				FTY (10^{-5} mol_{CO}/ g_{Fe} s)
			C_1	C_2	C_3	C_4	C_2-C_4	C_{5+}	C_1-C_4	$C_2^=$-$C_4^=$	C_2	C_3	C_4	C_2-C_4	
330	14.5	48.5	6.7	17.2	20.2	13.2	50.6	42.7	57.3	46.9	10.9	15.1	11.9	12.6	1.2
340	38.3	48.0	7.7	19.2	22.3	14.6	56.1	36.2	63.8	52.1	9.8	17.7	13.0	13.0	3.3
350	64.2	51.7	8.0	18.1	20.3	11.9	50.3	41.7	58.3	46.4	8.0	17.7	13.8	11.9	5.5
360	81.8	47.7	7.9	16.3	18.0	10.1	44.4	47.7	52.3	40.6	6.6	17.2	12.7	10.5	7.0
370	90.0	46.3	8.8	16.3	16.7	8.4	41.4	49.8	50.2	37.3	5.6	16.2	12.5	9.2	7.7

* Catalyst N1: 15%Fe/2%K_2O/83%($ZnAl_2O_4$•$1Al_2O_3$). P = 2.0 MPa. SV = 1500 mL•g cat^{-1}•h^{-1}. H_2/CO/Ar = 45/45/10. Iron time yield (FTY) represents moles of CO converted to hydrocarbons per unit of Fe per second.

Supporting Information Table S4. Catalytic performance of catalyst N5 *

T /°C	X_{CO} /%	S_{CO2} /%	hydrocarbon selectivity/% (C mole%)								olefin/paraffin ratio (O/P)				FTY (10^{-5} mol_{CO}/ g_{Fe} s)
			C_1	C_2	C_3	C_4	C_2-C_4	C_{5+}	C_1-C_4	$C_2^=$-$C_4^=$	C_2	C_3	C_4	C_2-C_4	
300	6.8	22.9	16.0	10.5	9.0	4.4	23.9	60.1	39.9	17.8	1.4	6.8	6.1	2.9	0.6
310	10.5	41.1	22.1	16.6	15.1	8.1	39.7	38.2	61.8	30.8	1.7	7.6	7.5	3.5	0.9
320	14.9	44.2	25.3	20.4	18.4	9.9	48.7	26.0	74.0	38.9	2.0	8.5	8.9	4.0	1.3
330	21.2	48.2	24.9	21.5	19.7	11.0	52.2	23.0	77.0	42.6	2.2	9.3	9.9	4.5	1.8
340	27.4	47.8	18.9	19.8	19.4	11.3	50.5	30.6	69.4	43.0	3.0	11.2	10.9	5.7	2.3
350	56.1	46.9	16.1	17.6	17.6	10.8	45.9	38.1	62.0	39.4	3.1	12.0	12.1	6.1	4.8
360	71.7	45.3	14.4	16.1	16.0	9.0	41.1	44.5	55.5	35.3	3.1	12.2	12.3	6.1	6.1
370	87.8	44.7	13.6	15.5	15.8	8.9	40.2	46.2	53.8	34.4	3.0	12.1	12.0	5.9	7.5

* Catalyst N5: 15%Fe/2%K_2O/83%($ZnAl_2O_4$•5Al_2O_3). P = 2.0 MPa. SV = 1500 mL•g cat^{-1}•h^{-1}. H_2/CO/Ar = 45/45/10. Iron time yield (FTY) represents moles of CO converted to hydrocarbons per unit of Fe per second.

Supporting Information Table S5. Catalytic performance of catalyst REF∞ *

T /°C	X_{CO} /%	S_{CO2} /%	hydrocarbon selectivity/% (C mole%)								olefin/paraffin ratio (O/P)				FTY (10^{-5} mol_{CO}/ g_{Fe} s)
			C_1	C_2	C_3	C_4	C_2-C_4	C_{5+}	C_1-C_4	$C_2^=$-$C_4^=$	C_2	C_3	C_4	C_2-C_4	
340	4.7	21.2	12.6	8.5	5.5	2.5	16.5	70.9	29.1	14.0	4.4	8.7	5.8	5.6	0.4
350	6.8	32.6	11.9	9.0	7.2	3.7	19.9	68.2	31.8	16.5	3.7	7.2	5.6	4.9	0.6
360	9.4	34.3	10.5	8.9	7.4	4.3	20.6	68.9	31.1	17.2	3.6	6.8	6.6	5.0	0.8
370	14.5	41.4	10.7	11.8	11.3	6.5	29.6	59.7	40.3	25.1	4.0	7.1	7.3	5.5	1.2
380	19.4	36.3	11.1	12.8	11.4	6.2	30.4	58.5	41.5	26.2	4.7	8.9	7.3	6.3	1.7
390	20.9	39.2	12.9	15.7	13.4	6.8	35.9	51.2	48.8	31.3	5.0	9.6	8.1	6.8	1.8
400	31.8	47.1	19.6	24.4	20.2	9.8	54.4	26.0	74.0	47.9	5.3	11.2	8.9	7.3	2.7
410	69.9	44.8	25.8	21.4	20.3	7.7	49.4	24.8	75.2	36.3	1.4	6.0	5.8	2.8	6.0
420	84.6	44.5	27.6	22.8	25.9	10.3	59.0	13.4	86.7	36.9	0.7	3.3	3.6	1.7	7.3

* Catalyst REF∞: 15%Fe/2%K_2O/83%(Al_2O_3). P = 2.0 MPa. SV = 1500 mL•g cat^{-1}•h^{-1}. H_2/CO/Ar = 45/45/10. Iron time yield (FTY) represents moles of CO converted to hydrocarbons per unit of Fe per second.

Supporting Information Table S6. Catalytic performance of catalyst REF0 *

T /°C	X_{CO} /%	S_{CO2} /%	hydrocarbon selectivity/% (C mole%)								olefin/paraffin ratio (O/P)				FTY (10^{-5} mol_{CO}/ g_{Fe} s)
			C_1	C_2	C_3	C_4	C_2-C_4	C_{5+}	C_1-C_4	$C_2^=$-$C_4^=$	C_2	C_3	C_4	C_2-C_4	
340	6.7	17.6	15.4	9.9	6.7	1.0	17.6	67.0	33.1	14.3	3.6	8.0	1.5	4.3	0.6
350	11.4	31.0	16.1	9.2	8.0	3.1	20.3	63.6	36.4	17.1	3.7	8.0	7.2	5.3	1.0
360	15.4	36.6	13.0	10.5	9.0	5.4	24.9	62.1	37.9	21.3	4.0	9.5	7.1	5.9	1.3
370	27.0	45.1	13.2	13.3	12.2	7.4	32.9	53.9	46.2	28.8	4.8	10.9	8.5	7.1	2.3
380	57.6	45.6	10.0	12.1	12.4	7.9	32.3	57.7	42.3	29.1	5.9	12.7	11.6	8.8	4.9
390	66.6	46.8	11.1	12.4	12.8	7.8	33.0	55.9	44.1	29.2	5.0	10.8	10.6	7.7	5.7
400	70.3	44.3	10.9	11.9	12.0	7.3	31.2	57.9	42.1	27.2	4.6	9.6	9.0	6.8	6.0
410	73.5	44.8	12.3	13.1	13.1	8.0	34.2	53.5	46.5	29.2	3.9	8.2	7.6	5.8	6.3

* Catalyst REF0: 15%Fe/2%K_2O/83%($ZnAl_2O_4$). P = 2.0 MPa. SV = 1500 mL•g cat^{-1}•h^{-1}. H_2/CO/Ar = 45/45/10. Iron time yield (FTY) represents moles of CO converted to hydrocarbons per unit of Fe per second.